

Altered chloride ion channel kinetics associated with the delta F508 cystic fibrosis mutation.

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Cystic fibrosis is associated with a defect in epithelial chloride ion transport which is caused by mutations in a membrane protein called CFTR (cystic fibrosis transmembrane conductance regulator). Heterologous expression of CFTR produces cyclicAMP-sensitive Cl(-)-channel activity. Deletion of phenylalanine at amino-acid position 508 in CFTR (delta F508 CFTR) is the most common mutation in cystic fibrosis. It has been proposed that this mutation prevents glycoprotein maturation and its transport to its normal cellular location. We have expressed both CFTR and delta F508 CFTR in Vero cells using recombinant vaccinia virus. Although far less delta F508 CFTR reached the plasma membrane than normal CFTR, sufficient delta F508 CFTR was expressed at the plasma membrane to permit functional analysis. delta F508 CFTR expression induced a reduced activity of the cAMP-activated Cl- channel, with conductance, anion selectivity and open-time kinetics similar to those of CFTR, but with much greater closed times, resulting in a large decrease of open probability. The delta F508 mutation thus seems to have two major consequences, an abnormal translocation of the CFTR protein which limits membrane insertion, and an abnormal function in mediating Cl- transport.

Cystic fibrosis. Part I. Frequency of the delta F508 mutation in South African families with cystic fibrosis.

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Cystic fibrosis (CF) is a common autosomal recessive disorder among people of European origin. With the localisation of the gene locus to chromosome 7q31 and the identification of closely linked polymorphic markers in 1985, it became possible to offer prenatal testing to couples at risk of having CF children, provided a live affected individual from that family was available for investigation. The CF gene, named CFTR, was cloned and sequenced in 1989 and the most common CF-causing mutation, delta F508, identified. A search for this mutation has been carried out in 81 South African white CF families of European origin. Using the polymerase chain reaction (PCR) technique the frequency of delta F508 was found to be 0.81. This mutation was not found in the 1 negroid and the 1 Indian CF family investigated.

An overview on chemical structures as Δ F508-CFTR correctors.

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Deletion of phenylalanine at position 508 (F508del) in the CFTR protein, is the most common mutation causing cystic fibrosis (CF). F508del causes misfolding and rapid degradation of CFTR protein a defect that can be targeted with pharmacological agents termed "correctors". Correctors belong to various chemical classes but are generally small molecules based on nitrogen sulfur or oxygen heterocycles. The mechanism of action of correctors is generally unknown but there is experimental evidence that some of them can directly act on mutant CFTR improving folding and stability. Here we overview the characteristics of the various F508del correctors described so far to obtain indications on key chemical structures and modifications that are required for mutant protein rescue.

Lipophilicity of the Cystic Fibrosis Drug, Ivacaftor (VX-770), and Its Destabilizing Effect on the Major CF-causing Mutation: F508del.

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Deletion of phenylalanine at position 508 (F508del) in cystic fibrosis transmembrane conductance regulator (CFTR) is the most common cystic fibrosis (CF)-causing mutation. Recently, ORKAMBI, a combination therapy that includes a corrector of the processing defect of F508del-CFTR (lumacaftor or VX-809) and a potentiator of channel activity (ivacaftor or VX-770), was approved for CF patients homozygous for this mutation. However, clinical studies revealed that the effect of ORKAMBI on lung function is modest and it was proposed that this modest effect relates to a negative impact of VX-770 on the stability of F508del-CFTR. In the current studies, we showed that this negative effect of VX-770 at 10 μ M correlated with its inhibitory effect on VX-809-mediated correction of the interface between the second membrane spanning domain and the first nucleotide binding domain bearing F508del. Interestingly, we found that VX-770 exerted a similar negative effect on the stability of other membrane localized solute carriers (SLC26A3, SLC26A9, and SLC6A14), suggesting that this negative effect is not specific for F508del-CFTR. We determined that the relative destabilizing effect of a panel of VX-770 derivatives on F508del-CFTR correlated with their predicted lipophilicity. Polarized total internal reflection fluorescence microscopy on a supported lipid bilayer model shows that VX-770, and not its less lipophilic derivative, increased the fluidity of and reorganized the membrane. In summary, our findings show that there is a potential for nonspecific effects of VX-770 on the lipid bilayer and suggest that this effect may account for its destabilizing effect on VX-809- rescued F508del-CFTR.

In Silico and In Vitro Evaluation of the Mechanism of Action of Three VX809-Based Hybrid Derivatives as Correctors of the F508del CFTR Protein.

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Cystic fibrosis (CF), the most common autosomal recessive fatal genetic disease in the Caucasian population, is caused by mutations in the gene encoding the cystic fibrosis transmembrane conductance regulator (CFTR), an anion channel that regulates salt and water transport across a variety of secretory epithelia. Deletion of phenylalanine at position 508, F508del, the most common CF-causing mutation, destabilises the CFTR protein, causing folding and trafficking defects that lead to a dramatic reduction in its functional expression. Small molecules called correctors have been developed to rescue processing-defective F508del CFTR. We have combined in silico and in vitro approaches to investigate the mechanism of action and potential as CFTR correctors of three hybrid derivatives (2a, 7a, and 7m) obtained by merging the amino-arylthiazole core with the benzodioxole carboxamide moiety characterising the corrector lumacaftor. Molecular modelling analyses suggested that the three hybrids interact with a putative region located at the MSD1/NBD1 interface. Biochemical analyses confirmed these results, showing that the three molecules affect the expression and stability of the F508del NBD1. Finally, the YFP assay was used to evaluate the influence of the three hybrid derivatives on F508del CFTR function, assessing that their effect is additive to that of the correctors VX661 and VX445. Our study shows that the development and testing of optimised compounds targeting different structural and functional defects of mutant CFTR is the best strategy to provide more effective correctors that could be used alone or in combination as a valuable therapeutic option to treat an even larger cohort of people affected by CF.

A Novel Mutation in Cystic Fibrosis Presenting as Pseudo Bartter Syndrome: A Case Report.

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Pseudo-Bartter's (PB) syndrome is characterized by hypokalemic metabolic alkalosis and failure to thrive which constitutes a rare but typical presentation of cystic fibrosis (CF) in children. The most common mutation of CF is F508del, due to loss of 3 base pairs, causing deletion of phenylalanine, at position 508. We present a case of CF presenting with failure to thrive, dehydration, PB syndrome associated with urosepsis and primo-colonization with *Escherichia coli* suggesting the role of epigenetic factors. The heterozygous state for Phe508del mutation in Exon 11 combination with Glu92Ala in Exon 4 resulted in epigenetic effect on atypical phenotype as PBS, a novel mutation identified in our case.

Partial Rescue of F508del-CFTR Stability and Trafficking Defects by Double Corrector Treatment.

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Deletion of phenylalanine at position 508 (F508del) in the CFTR chloride channel is the most frequent mutation in cystic fibrosis (CF) patients. F508del impairs the stability and folding of the CFTR protein, thus resulting in mistrafficking and premature degradation. F508del-CFTR defects can be overcome with small molecules termed correctors. We investigated the efficacy and properties of VX-445, a newly developed corrector, which is one of the three active principles present in a drug (Trikafta®/Kaftrio®) recently approved for the treatment of CF patients with F508del mutation. We found that VX-445, particularly in combination with type I (VX-809, VX-661) and type II (corr-4a) correctors, elicits a large rescue of F508del-CFTR function. In particular, in primary bronchial epithelial cells of CF patients, the maximal rescue obtained with corrector combinations including VX-445 was close to 60-70% of CFTR function in non-CF cells. Despite this high efficacy, analysis of ubiquitylation, resistance to thermoaggregation, protein half-life, and subcellular localization revealed that corrector combinations did not fully normalize F508del-CFTR behavior. Our study indicates that it is still possible to further improve mutant CFTR rescue with the development of corrector combinations having maximal effects on mutant CFTR structural and functional properties.

The $\Delta F508$ mutation causes CFTR misprocessing and cystic fibrosis-like disease in pigs.

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Cystic fibrosis (CF) is an autosomal recessive disease caused by mutations in the gene encoding the cystic fibrosis transmembrane conductance regulator (CFTR) anion channel. The most common CF-associated mutation is $\Delta F508$, which deletes a phenylalanine in position 508. In vitro studies indicate that the resultant protein, CFTR- $\Delta F508$, is misprocessed, although the in vivo consequences of this mutation remain uncertain. To better understand the effects of the $\Delta F508$ mutation in vivo, we produced CFTR($\Delta F508/\Delta F508$) pigs. Our biochemical, immunocytochemical, and electrophysiological data on CFTR- $\Delta F508$ in newborn pigs paralleled in vitro predictions. They also indicated that CFTR($\Delta F508/\Delta F508$)

airway epithelia retain a small residual CFTR conductance, with maximal stimulation producing ~6% of wild-type function. Cyclic adenosine 3',5'-monophosphate (cAMP) agonists were less potent at stimulating current in CFTR(Δ)(F508/)(Δ)(F508) epithelia, suggesting that quantitative tests of maximal anion current may overestimate transport under physiological conditions. Despite residual CFTR function, four older CFTR(Δ F508/ Δ F508) pigs developed lung disease similar to human CF. These results suggest that this limited CFTR activity is insufficient to prevent lung or gastrointestinal disease in CF pigs. These data also suggest that studies of recombinant CFTR- Δ F508 misprocessing predict in vivo behavior, which validates its use in biochemical and drug discovery experiments. These findings help elucidate the molecular pathogenesis of the common CF mutation and will guide strategies for developing new therapeutics.

Buserelin alleviates chloride transport defect in human cystic fibrosis nasal epithelial cells.

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Cystic fibrosis (CF) is the most common autosomal recessive disease in Caucasians caused by mutations in the gene encoding the Cystic Fibrosis Transmembrane conductance Regulator (CFTR) chloride (Cl⁻) channel regulated by protein kinases, phosphatases, divalent cations and by protein-protein interactions. Among protein-protein interactions, we previously showed that Annexin A5 (AnxA5) binds to CFTR and is involved in the channel localization within membranes and in its Cl⁻ channel function. The deletion of phenylalanine at position 508 (F508del) is the most common mutation in CF which leads to an altered protein (F508del-CFTR) folding with a nascent protein retained within the ER and is quickly degraded. We previously showed that AnxA5 binds to F508del-CFTR and that its increased expression due to a Gonadoliberein (GnRH) augments Cl⁻ efflux in cells expressing F508del-CFTR. The aim of the present work was to use the GnRH analog buserelin which is already used in medicine. Human nasal epithelial cells from controls and CF patients (F508del/F508del) were treated with buserelin and we show here that the treatment alleviates Cl⁻ channel defects in CF cells. Using proteomics we highlighted some proteins explaining this result. Finally, we propose that buserelin is a potential new pharmaceutical compound that can be used in CF and that bronchus can be targeted since we show here that they express GnRH-R.

Potential of Δ F508- and G551D-CFTR-Mediated Cl⁻ Current by Novel Hydroxypyrazolines.

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The most common mutation of CFTR, affecting approximately 90% of CF patients, is a deletion of phenylalanine at position 508 (F508del, Δ F508). Misfolding of Δ F508-CFTR impairs both its trafficking to the plasma membrane and its chloride channel activity. To identify small molecules that can restore channel activity of Δ F508-CFTR, we synthesized and evaluated eighteen novel hydroxypyrazoline analogues as CFTR potentiators. To elucidate potentiation activities of hydroxypyrazolines for Δ F508-CFTR, CFTR activity was measured using a halide-sensitive YFP assay, Ussing chamber assay and patch-clamp technique. Compounds 7p, 7q and 7r exhibited excellent potentiation with EC₅₀ value <10 μ M. Among the compounds, 7q (a novel CFTR potentiator, CP7q) showed the highest potentiation activity with EC₅₀ values of 0.88 ± 0.11 and 4.45 ± 0.31 μ M for wild-type and Δ F508-CFTR, respectively. In addition, CP7q significantly potentiated chloride conductance of G551D-CFTR, a CFTR gating mutant; its

maximal potentiation activity was 1.9 fold higher than the well-known CFTR potentiator genistein. Combination treatment with CP7q and VX-809, a corrector of $\Delta F508$ -CFTR, significantly enhanced functional rescue of $\Delta F508$ -CFTR compared with VX-809 alone. CP7q did not alter the cytosolic cAMP level and showed no cytotoxicity at the concentration showing maximum efficacy. The hydroxypyrazolines may be potential development candidates for drug therapy of cystic fibrosis.
